

CYBER DETECTIVE CHALLENGE (INTRODUCTION TO ETHICAL HACKING)

1. Project Title

Cyber Detective Challenge – Learn Ethical Hacking Basics Through a Web App

2. Project Objectives

By completing this project, students will be able to:

- Understand the basics of ethical hacking and cybersecurity.
- Identify weak and strong passwords.
- Recognize simple phishing attempts.
- Understand how encryption (hidden messages) works.
- Use basic HTML, CSS, and JavaScript.
- Build a simple interactive web application.
- Develop problem-solving and logical thinking skills.

3. Project Description

Students will build a **Cyber Detective Web App** that simulates real cybersecurity tasks in a fun and safe way.

The web app includes:

- Password strength checker
- Secret message decoder
- Cybersecurity quiz
- Certificate generator

The goal is to complete all missions and become a **Junior Cyber Detective**.

4. Instructions for Students

Step 1: Setup Project Folder

Create a folder named:

cyber-detective-app

Inside it, create three files:

- `index.html`
- `style.css`
- `script.js`

Step 2: Build the Web App

You will write code in three parts:

- HTML → Structure of the app
- CSS → Design and styling
- JavaScript → Logic and interactivity

5. CODE SECTION

A) index.html (Main Structure)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Cyber Detective Challenge</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <div class="container">

    <h1>🕵️ Cyber Detective Challenge</h1>
    <p>Complete all missions and become a Junior Cyber Detective!</p>

    <!-- PASSWORD CHECKER -->
    <div class="card">
      <h2>Mission 1: Password Guardian</h2>

      <input type="text" id="password" placeholder="Enter password">
      <button onclick="checkPassword()">Check Password</button>

      <p id="passwordResult"></p>
    </div>

    <!-- SECRET MESSAGE -->
    <div class="card">
```

```
<h2>Mission 2: Secret Message</h2>
```

```
<button onclick="showMessage()">Reveal Message</button>
```

```
<p id="secretMessage"></p>
```

```
</div>
```

```
<!-- QUIZ -->
```

```
<div class="card">
```

```
<h2>Mission 3: Cyber Quiz</h2>
```

```
<p>1. Should you share your password?</p>
```

```
<input type="radio" name="q1" value="no"> No
```

```
<input type="radio" name="q1" value="yes"> Yes
```

```
<p>2. What is phishing?</p>
```

```
<input type="radio" name="q2" value="fake"> Fake message stealing data
```

```
<input type="radio" name="q2" value="fish"> Fishing activity
```

```
<p>3. Is "123456" strong?</p>
```

```
<input type="radio" name="q3" value="no"> No
```

```
<input type="radio" name="q3" value="yes"> Yes
```

```
<br><br>
```

```
<button onclick="checkQuiz()">Submit Quiz</button>
```

```
<p id="quizResult"></p>
```

```
</div>
```

```
<!-- CERTIFICATE -->
```

```
<div class="card">
```

```
<h2>🏆 Certificate Generator</h2>
```

```
<input type="text" id="studentName" placeholder="Enter your name">
```

```
<button onclick="generateCertificate()">Generate Certificate</button>
```

```
<div id="certificate"></div>
```

```
</div>
```

```
</div>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

B) style.css (Design)

```
body {
  font-family: Arial;
  background: #0f172a;
  color: white;
  padding: 20px;
}

.container {
  max-width: 900px;
  margin: auto;
}

h1 {
  text-align: center;
  color: #38bdf8;
}

.card {
  background: #1e293b;
  padding: 20px;
  margin: 15px 0;
  border-radius: 10px;
}

input {
  padding: 10px;
  margin: 5px 0;
  width: 100%;
}

button {
  padding: 10px;
  background: #38bdf8;
  border: none;
  cursor: pointer;
  font-weight: bold;
}

#certificate {
  display: none;
  margin-top: 20px;
  padding: 15px;
  border: 2px dashed gold;
}
```

C) script.js (Functionality)

```
// PASSWORD CHECKER
```

```
function checkPassword() {
```

```
    let password = document.getElementById("password").value;
```

```
    let result = document.getElementById("passwordResult");
```

```
    if (password.length >= 8) {
```

```
        result.innerHTML = "Strong Password ";
```

```
    } else {
```

```
        result.innerHTML = "Weak Password  (Use at least 8 characters)";
```

```
    }
```

```
}
```

```
// SECRET MESSAGE
```

```
function showMessage() {
```

```
    document.getElementById("secretMessage").innerHTML =
```

```
    "🔒 Welcome Cyber Detective!";
```

```
}
```

```
// QUIZ SYSTEM
```

```
function checkQuiz() {
```

```
    let score = 0;
```

```
    let q1 = document.querySelector('input[name="q1"]:checked');
```

```
    let q2 = document.querySelector('input[name="q2"]:checked');
```

```
    let q3 = document.querySelector('input[name="q3"]:checked');
```

```
    if (q1 && q1.value === "no") score++;
```

```
    if (q2 && q2.value === "fake") score++;
```

```
    if (q3 && q3.value === "no") score++;
```

```
    document.getElementById("quizResult").innerHTML =
```

```
    "Your Score: " + score + "/3";
```

```
}
```

```
// CERTIFICATE GENERATOR
```

```
function generateCertificate() {
```

```
    let name = document.getElementById("studentName").value;
```

```
    if (name === "") {
```

```
        alert("Please enter your name");
```

```
        return;
```

```
    }
```

```

let cert = document.getElementById("certificate");
cert.style.display = "block";

cert.innerHTML = `
  <h2>🏆 Certificate of Achievement</h2>
  <p>This certifies that</p>
  <h1>${name}</h1>
  <p>has completed the Cyber Detective Challenge</p>
  <h3>Junior Cyber Detective</h3>
  `;
}

```

6. Learning Outcomes

By completing this project, students will:

- Build their first interactive web application
- Understand basic cybersecurity concepts
- Learn how ethical hackers think
- Apply programming logic using JavaScript
- Improve creativity and problem-solving skills

7. Assessment Criteria (100 Marks)

Component	Marks
HTML Structure	25
CSS Design	20
JavaScript Functionality	30
Cybersecurity Understanding	15
Creativity & Completion	10

8. Bonus Challenge (Optional)

Improve your app by adding:

- Sound effects for correct answers
- Countdown timer for quiz
- Score tracking system
- Animated certificate (confetti 🎉)
- Dark/light mode toggle

9. Final Outcome

At the end of the project, students will have built a:



Cyber Detective Web App

that teaches cybersecurity through interactive learning and simple coding.